Name: **CSE 450, Fall 2026 – Quiz 1B**

Assume lexer-style regular expression syntax as used in Emplex or flex (and taught in class). Unmatched input returns the next single character. In Questions 4–10, a match must cover the entire string. Choose one answer per question. Each question is worth 1 point, for 10 points total.

For the first three questions, consider these lexer rules in the order shown:

RETURN	`"return"`
CAPS	`[A-Z]+`
NUM	`[0-9]+`
ID	`[a-zA-Z_][a-zA-Z0-9_]*`

1. What would the lexer return for the input `returnValue`?

A. A single RETURN token
B. A single ID token
C. A single CAPS token
D. A single NUM token
E. A RETURN and then an ID token
F. A RETURN and then a CAPS token

2. What would the same lexer return for this input: `27+`

A. A single NUM token
B. Only the character '+'
C. A NUM and then an CAPS token
D. A NUM token and then the character '+'
E. The character '+' and then a NUM token
F. A single ID token

3. Suppose the ID rule is moved immediately before CAPS, leaving RETURN first. How would this affect which token types can be returned?

A. CAPS would never be returned, but RETURN could still be returned.
B. RETURN would never be returned, but CAPS could still be returned.
C. Neither CAPS nor RETURN could ever be returned.
D. Nothing would change, because rule order never affects the token type.
E. ID would never be returned, because CAPS has the more specific pattern.

4. How many words are in the regular language represented by $\mathcal{L}$(`[abcd][xyz]`) ?

A. 6 strings
B. 7 strings
C. 9 strings
D. 12 strings
E. 18 strings
F. 27 strings

5. Which regular expression matches exactly the binary strings with exactly two 1s? It should match `11`, `101`, and `0010010`, but reject `000`, `10`, and `111`. Any number of 0s may appear anywhere in the sequence.

A. `[01]*1[01]*1[01]*`

B. `0+10+10+`

C. `0*10*10*`

D. `0*1?0*1?0*`

E. `(0*1)*0*`

F. `0*110*`

6. You want to match the three character literal string `"a.b"` and nothing else. Which regular expression will NOT restrict matches to exactly `a.b`?

A. `"a.b"`

B. `a\.b`

C. `a[.]b`

D. `a.b`

E. `a"."b`

F. `[a][.][b]`

7. How do the following regular expressions compare to each other?

A: `(a|b)*` **B: `a+b*`**

Hint: Consider the empty string and the strings `a`, `ab`, and `ba`.

A. Every string matched by A is matched by B, but B matches additional strings.

B. Every string matched by B is matched by A, but A matches additional strings.

C. A and B match exactly the same strings.

D. A and B have no matching strings in common.

E. They share some matches, but each also matches strings the other does not.

8. You want to match one or more copies of the two-character sequence `ab`, such as `ab` and `abab`. A student proposes `[ab]+`. Will this work in all cases?

A. Yes. The brackets keep a and b together, and + repeats the pair.

B. No. The + allows zero copies, so it would include the empty string.

C. No. Each repetition can choose a or b independently, so it also accepts aa and ba.

D. No. A character class can only match once, so it cannot match abab.

E. No. The brackets require a literal [before the letters and a literal] after them.

F. No. The + applies only to b, so the pattern matches one a followed by one or more b's.

9. Which description best matches the language of `[a-z][0-9]([a-z][0-9])*`?

A. All nonempty strings containing only lowercase letters and digits, in any order.

B. A single lowercase letter followed by one or more digits, such as a7 or b123.

C. One fixed letter-digit pair appearing one or more times; a1a1 matches, but a1b2 does not.

D. All strings of exactly two characters: one lowercase letter followed by one digit.

E. One or more letter-digit pairs, with each pair chosen independently; a1 and a1b2 match.

10. A command name consists of `get` or `set`, followed by one or more digits. Which regular expression matches exactly these command names? It should accept `get7` and `set204`, but reject `get`, `set`, and `getset3`.

A. `(get|set)[0-9]+`

B. `get|set[0-9]+`

C. `(get|set)[0-9]*`

D. `(get|set[0-9])+`

E. `(get|set)+[0-9]+`

EXTRA CREDIT:

Write a regular expression that allows for matched parentheses with a single level of nesting. The regular expression should allow `()`, `()()`, `(())`, or `(()())()`, but reject `(()`, `)(`, or `())` for being unbalanced, and reject `((()))` for being nested by more than a single level.